

A Machine-Verified Disproof of TLMC Conjecture #1260: Substitutive Factor Complexity Is Not Bounded by $n + 2$

Batch 12 solution submission

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Abstract

We disprove a conjecture of the Last Math Competition asserting that within the purely substitutive class the factor complexity satisfies $p(n) \leq n + 2$, with $p(n) = n + 1$ only for Sturmian words. The Thue–Morse word — the fixed point of the binary substitution $0 \mapsto 01$, $1 \mapsto 10$, plainly purely substitutive — has $p(3) = 6 > 5 = 3 + 2$: its length-3 factors are exactly 001, 010, 011, 100, 101, 110. The Lean 4 formalization builds the substitution on lists, proves the iterate-prefix property (via the homomorphism $\sigma(a++b) = \sigma a ++ \sigma b$), realizes the fixed point by the bit recursion $\text{tm}(2k) = \text{tm}(k)$, $\text{tm}(2k+1) = \neg \text{tm}(k)$, identifies it with the fifth iterate on 32 symbols, and counts the distinct factors by kernel computation. Zero errors/warnings; the decisive theorems depend on *no* axioms.

1 The conjecture

Conjecture 1.1 (TLMC-00000001260). The optimal upper bound for factor complexity within the purely substitutive class is $p(n) \leq n + 2$, attained by three-letter substitution families beyond the Tribonacci word; $p(n) = n + 1$ occurs only for Sturmian words.

2 The disproof

Theorem 2.1. *Conjecture 1.1 is **false** in its main clause: the Thue–Morse word w (fixed point of $0 \mapsto 01$, $1 \mapsto 10$) is purely substitutive and*

$$p_w(3) = 6 > 5 = 3 + 2.$$

Proof. The length-3 factors of $w = 0110100110010110 \cdots$ are exactly $\{001, 010, 011, 100, 101, 110\}$ (the cubes 000 and 111 never occur, as w is overlap-free; all six listed factors occur within the first 32 symbols = the fifth substitution iterate). Hence $p_w(3) = 6$. $\square$

Remark. The Sturmian clause ($p(n) = n + 1$ only for Sturmian) is a classical theorem (Coven–Hedlund type) and is not disputed. The failure is the claimed *upper bound*: substitutive words can have complexity growing with an arbitrarily large linear slope (Thue–Morse has $p(n) \sim \frac{10}{3}n$; the claim $p(n) \leq n + 2$ already fails at $n = 3$).

3 Machine verification

`Batch12.lean` (namespace `TLMCBatch12`): `sigma` and `sigma_append` (list homomorphism), `sigma_iter_append` and `t_prefix` (iterates form a chain of prefixes, so the direct limit — the fixed point — exists), the fixed point realized by `tm/tmA` (fuel-structural bit recursion) with `tm_eq_iterate` (identification on 32 symbols, kernel-checked), and `tm_factors_3` (exactly 6 distinct length-3 factors) with `t1m1260_false`. Compilation exits 0 with zero errors and warnings; `#print axioms`: the two decisive theorems depend on *no* axioms, the prefix lemmas at most `[propext, Quot.sound]`; no `sorry`.